RHEL 9 vs Ubuntu 24.04 LTS — Detailed Comparison

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1. Introduction

This document provides a detailed technical and operational comparison between **Red Hat Enterprise Linux 9 (RHEL 9)** and **Ubuntu 24.04 LTS ("Noble Numbat")**, the two most widely deployed enterprise Linux distributions. It is intended for system administrators, DevOps engineers, and architects evaluating which OS best fits their workload, compliance, budget, and support requirements.

2. Overview & Lineage

	RHEL 9	Ubuntu 24.04 LTS
Vendor	Red Hat (IBM subsidiary)	Canonical Ltd.
Upstream lineage	Built from CentOS Stream 9 → Fedora origins	Built from Debian "Testing" (sid) snapshot
First released	May 17, 2022	April 25, 2024
Current point release (late 2025)	9.5 / 9.6	24.04.2 / 24.04.3
Philosophy	Stability, long-term binary compatibility, certification-first	Rapid feature adoption, broad cloud/community reach, free-first
Governance	Corporate (Red Hat), community input via Fedora/CentOS Stream	Corporate (Canonical) + community via Debian/Ubuntu dev boards

3. Release & Support Lifecycle

	RHEL 9	Ubuntu 24.04 LTS
Full (Production Phase 1) support	5 years → ends May 2027	5 years (standard) → ends April 2029
Extended support	Phase 2 (reduced) to May 2029; ELS to May 2032	ESM via Ubuntu Pro to April 2034
Point releases	Roughly every 6 months (9.0 → 9.6 ...)	~Every 3-6 months (.1, .2, .3)
In-place major upgrade	`leapp` (RHEL 8→9, 9→10)	`do-release-upgrade`
EOL notification tooling	Red Hat Insights, subscription-manager	`pro status` , `do-release-upgrade -c`

> **Note:** Non-LTS Ubuntu releases (e.g., 24.10, 25.04) only get 9 months of support and are not suited for production servers — always use LTS (`.04` releases of even years) for enterprise use.

4. Supported Architectures

Architecture	RHEL 9	Ubuntu 24.04
x86_64	✅	✅
aarch64 (ARM64)	✅	✅
ppc64le (IBM Power)	✅	✅
s390x (IBM Z/mainframe)	✅	✅ (server)
RISC-V	❌	✅ (community/experimental)
32-bit x86	❌ (dropped)	❌ (dropped since 19.10)

5. Installation & Default Partitioning

	RHEL 9	Ubuntu 24.04
Installer	Anaconda (graphical/text/kickstart)	Subiquity (server) / Ubiquity-successor (desktop)
Automated install	Kickstart (`.ks` files)	Autoinstall (cloud-init-based YAML)
Default partition layout	LVM on top of single `/boot` + `/` (XFS)	Single ext4 partition, LVM optional
Default swap	Swap partition or swap file depending on install type	Swap file (not partition) by default since 22.04+
Minimal install footprint	~1.2–1.5 GB	~1.1 GB

6. Package Management (Deep Dive)

	RHEL 9	Ubuntu 24.04
Low-level tool	`rpm`	`dpkg`
High-level tool	`dnf` (replaces yum)	`apt` / `apt-get` / `aptitude`
Repo config location	`/etc/yum.repos.d/*.repo`	`/etc/apt/sources.list` + `sources.list.d/`
Package signing	GPG via `rpm --import`	GPG keys in `/etc/apt/trusted.gpg.d/`
Metadata caching	`dnf makecache`	`apt update`
Modular/versioned software	**AppStream module streams** (e.g., `dnf module enable postgresql:15`)	No native module system — PPAs or Snap channels
Universal/sandboxed apps	Flatpak (opt-in, Flathub not default)	**Snap** (pre-installed; Firefox etc. ship as snaps by default)
Offline/local repo creation	`createrepo_c` + `reposync`	`apt-mirror`, `dpkg-scanpackages`
Downgrade package	`dnf downgrade <pkg>`	`apt install <pkg>=<version>`
Clean unused packages	`dnf autoremove`	`apt autoremove`

> **Operational note:** Ubuntu's push toward Snap (Firefox, Chromium ship only as snaps) is a frequent friction point for admins who prefer deb-only systems — can be blacklisted post-install if desired.

7. Filesystem & Storage

	RHEL 9	Ubuntu 24.04
Default root filesystem	**XFS**	**ext4**
Alternative FS support	ext4, Btrfs (tech preview), Stratis	ZFS (officially supported root-on-ZFS), Btrfs, XFS
LVM support	Default-on, full tooling	Supported, optional at install
Volume mgmt abstraction	**Stratis** (pools, thin provisioning)	**ZFS** (pools, snapshots, native encryption)
Snapshot support	LVM snapshots, Btrfs (unsupported)	ZFS snapshots, LVM snapshots, Timeshift (desktop)
Max supported FS size	XFS: 500 TB (supported limit)	ext4: 1 EB theoretical; ZFS far larger

8. Kernel, Toolchain & Boot Process

	RHEL 9	Ubuntu 24.04
Kernel (GA)	5.14, heavily backported	6.8 mainline, HWE updates available mid-cycle
Kernel update cadence	Security/bugfix backports only	HWE stack can bump kernel minor versions during LTS life
Bootloader	GRUB2 (Boot Loader Specification)	GRUB2 (classic grub.cfg via update-grub)
initramfs tool	`dracut`	`initramfs-tools`
Default compiler	GCC 11 (gcc-toolset for newer)	GCC 13
glibc	2.34	2.39
Python (system default)	3.9 (3.11/3.12 via module)	3.12
Secure Boot	Supported (Red Hat signed shim)	Supported (Canonical signed shim)

9. Security

	RHEL 9	Ubuntu 24.04
Mandatory Access Control	**SELinux** — label-based, enforcing by default, very granular	**AppArmor** — path-based, enabled by default, simpler profile model
Firewall front-end	`firewalld` (zone-based, nftables backend)	`ufw` (simple rules, iptables/nftables backend)
System-wide crypto policy	`update-crypto-policies` (DEFAULT/LEGACY/FUTURE/FIPS) — applies globally	No equivalent single switch — per-app config
FIPS 140-3 validated modules	Yes, first-class (`fips-mode-setup --enable`)	Available via Ubuntu Pro add-on
CVE/patch tracking	Red Hat CVE DB + `dnf updateinfo`	Ubuntu Security Notices (USN) + `apt list --upgradable`
Gov/compliance certs	Common Criteria, FIPS 140-3, DISA STIG — maintained by Red Hat	CIS Benchmarks, FIPS via Pro, fewer out-of-box STIG profiles
Auditing	`auditd` integrated with SELinux events	`auditd` available, less SELinux-style correlation
Default root login	Disabled; sudo-based	Disabled; sudo-based (install-time user auto-added to `sudo`)

10. User & Group Management

	RHEL 9	Ubuntu 24.04
Admin group	`wheel`	`sudo`
Default `useradd` behavior	Creates private group per user	Creates private group per user
Password aging defaults	`/etc/login.defs` ~99999 days unless hardened	`/etc/login.defs` similar, often tightened via CIS tooling
Central identity integration	SSSD, tightly integrated with FreeIPA/AD	SSSD or `realmd`, equally supported

11. Networking

	RHEL 9	Ubuntu 24.04
Primary network manager	NetworkManager (`nmcli`, `nmtui`)	Netplan (YAML) → NetworkManager/systemd-networkd
Config file location	`/etc/NetworkManager/system-connections/`	`/etc/netplan/*.yaml`
DNS resolution	`systemd-resolved` optional	`systemd-resolved` enabled by default
Static IP example	`nmcli con mod eth0 ipv4.addresses 192.168.1.10/24 ipv4.method manual`	Edit `/etc/netplan/*.yaml`, then `netplan apply`
Hostname tool	`hostnamectl`	`hostnamectl`

12. Logging & Monitoring

	RHEL 9	Ubuntu 24.04
Journal	`systemd-journald` / `journalctl`	`systemd-journald` / `journalctl`
Traditional syslog	`rsyslog` → `/var/log/messages`	`rsyslog` → `/var/log/syslog`
Log rotation	`logrotate`, `/etc/logrotate.d/`	`logrotate`, `/etc/logrotate.d/`
Built-in perf tooling	`sos report`, PCP, Red Hat Insights	`sosreport` via apt; Canonical Landscape for fleets
Default monitoring agent	`insights-client`	`ua` / `pro` client, Livepatch

13. Process & Service Management

	RHEL 9	Ubuntu 24.04
Init/service manager	systemd	systemd
Custom unit file location	`/etc/systemd/system/`	`/etc/systemd/system/`
Service enable/start	`systemctl enable --now <svc>`	`systemctl enable --now <svc>`
Default target	`multi-user.target` / `graphical.target`	`multi-user.target` / `graphical.target`

Service management commands are identical since both use systemd — this is one of the biggest convergence points between the two OSes.

14. Containers & Virtualization

	RHEL 9	Ubuntu 24.04
Default container engine	**Podman** (daemonless, rootless-first)	**Docker**/ `containerd` common; Podman via apt/snap
First-party orchestration	OpenShift (commercial), MicroShift (edge)	MicroK8s, Charmed Kubernetes
Hypervisor	KVM/QEMU via libvirt/virt-manager, oVirt/RHV at scale	KVM/QEMU via libvirt/virt-manager, MAAS for bare-metal
Cloud-init support	Yes	Yes (Ubuntu maintains cloud-init upstream)
VM image focus	qcow2, RHEL cloud/KVM guest images	qcow2, OVA; most widely used public cloud base image

15. High Availability & Clustering

	RHEL 9	Ubuntu 24.04
Cluster stack	Pacemaker + Corosync, Red Hat supported	Pacemaker + Corosync via apt, community-supported
Resource/fencing agents	Extensive, certified library	Available, less vendor-certified
Shared storage clustering	**GFS2**	No direct equivalent — typically Ceph/GlusterFS
Load balancing tooling	HAProxy, Keepalived (EPEL/subscription)	HAProxy, Keepalived (native in main repos)

16. Performance Tuning

	RHEL 9	Ubuntu 24.04
Tuning framework	`tuned` daemon with profiles	No built-in equivalent by default; manual sysctl or install `tuned`
CPU governor tools	`tuned-adm`, `cpupower`	`cpupower`, `tlp` (laptops/desktops)
NUMA tooling	`numactl`, `numad`	`numactl` (numad not default)
I/O scheduler defaults	`mq-deadline`	`mq-deadline` / `none` (NVMe)

17. Licensing & Cost

	RHEL 9	Ubuntu 24.04
Base OS acquisition	Free download, but **updates require active subscription** (free dev tier capped at 16 systems)	**Completely free**, including security updates, forever
Support tiers	Self-Support, Standard, Premium (24/7) — priced per socket-pair/VM	Free community; **Ubuntu Pro** free ≤5 machines, paid per-node for business (ESM, FIPS, CIS, Livepatch)
Cloud marketplace pricing	Often an **hourly RHEL surcharge** on top of infra cost	Images typically free — pay only for compute
ISV certification impact	Often first/exclusive certification target (SAP, Oracle)	Growing fast, especially cloud-native ISVs

18. Ecosystem & Vendor Support

	RHEL 9	Ubuntu 24.04
Hardware vendor certification	Deep OEM matrix (Dell, HPE, Lenovo, Cisco UCS)	Broadly certified; strong on cloud VM/ARM/IoT
ISV software certification	SAP HANA, Oracle DB, IBM Db2 often primary target	Expanding fast in K8s/cloud-native ISV ecosystem
Cloud marketplace presence	Available on all major clouds, "enterprise" tier	Dominant base image across AWS, Azure, GCP, OCI
Community size/forums	Large enterprise community, subscription-gated KB	Very large open community (Ask Ubuntu, Launchpad)

19. Pros & Cons Summary

RHEL 9

Pros:

- Rock-solid ABI/API stability for the full lifecycle — ideal for ISV-certified workloads.
- SELinux + system-wide crypto policies give best-in-class out-of-box security posture.
- Strong government/regulated-industry certification coverage (FIPS, STIG, Common Criteria).
- Vendor-backed enterprise SLA and 24/7 support options.
- `leapp` provides a well-tested in-place major upgrade path.

Cons:

- Requires paid subscription for ongoing patching in production (cost scales with fleet size).
- Older default package versions (stability-first) can lag behind latest upstream features.
- Smaller "free" footprint — subscription management adds operational overhead.

Ubuntu 24.04 LTS

Pros:

- Completely free, including security patching, for the full support window.
- Newer kernel/toolchain/library versions at release — better for cutting-edge hardware/dev tooling.
- Dominant presence in public cloud and container ecosystems — easiest to find community guides/images.
- `cloud-init` and Netplan designed cloud-first, simplifying automated provisioning.

Cons:

- Snap-first strategy for core apps is controversial among admins who prefer pure deb packaging.
- AppArmor is generally considered less granular/powerful than SELinux for fine-grained MAC policies.
- Fewer out-of-box regulated-industry certifications compared to RHEL (Ubuntu Pro narrows this gap).
- HA/clustering (Pacemaker/GFS2-style stack) is less of a first-party focus than on RHEL.

20. Decision Matrix — Which to Choose

Scenario	Recommended OS	Why
Regulated industry requiring FIPS/STIG	**RHEL 9**	Built-in certified crypto policies & compliance tooling
Running SAP HANA, Oracle DB, or ISV-certified software	**RHEL 9**	Primary/first certification target for most enterprise ISVs
Cloud-native microservices / Kubernetes-heavy	**Ubuntu 24.04**	Dominant cloud image, large K8s ecosystem alignment, free
Tight budget / large horizontal server fleet	**Ubuntu 24.04**	No subscription cost for patching at scale
Need vendor SLA + 24/7 support contract	**RHEL 9**	Red Hat subscription includes enterprise support tiers
Dev/test environments, CI/CD runners	**Ubuntu 24.04**	Fast setup, free, huge community tooling/documentation
On-prem HA cluster with shared storage (GFS2)	**RHEL 9**	Native Pacemaker/Corosync/GFS2 integration & certification
IoT / edge devices	**Ubuntu Core**	Purpose-built minimal, snap-based, OTA-updatable edge OS

21. Migration Considerations

- **Packages**: RPM ⇄ DEB are not interchangeable; `alien` conversion exists but is unreliable for production — rebuild/re-source packages natively for the target OS.
- **Security policy**: SELinux policies and AppArmor profiles must be redesigned, not ported, when switching OS.
- **Automation**: Ansible/Terraform playbooks referencing `dnf` / `apt` module names or OS-specific facts (`ansible_os_family`) need conditional branches.
- **Licensing**: Moving RHEL → Ubuntu removes subscription costs but also removes vendor SLA — assess support risk tolerance.
- **Validate ISV certification** of all critical enterprise software on the target OS before migrating production workloads.
- **Backup/restore data layer** (DB dumps, config) rather than attempting filesystem-level OS conversion — treat as fresh install + data migration.

22. Directory & Config File Locations

Purpose	RHEL 9	Ubuntu 24.04
Package repo config	`/etc/yum.repos.d/`	`/etc/apt/sources.list(.d/)`
Network config	`/etc/NetworkManager/system-connections/`	`/etc/netplan/`
Firewall rules	`/etc/firewalld/`	`/etc/ufw/` , `/etc/iptables/`
Sudo rules	`/etc/sudoers` , `/etc/sudoers.d/`	`/etc/sudoers` , `/etc/sudoers.d/`
SELinux/AppArmor policy	`/etc/selinux/` , `semanage`	`/etc/apparmor.d/`
systemd unit overrides	`/etc/systemd/system/`	`/etc/systemd/system/`
Cron jobs	`/etc/cron.d/` , `/var/spool/cron/`	`/etc/cron.d/` , `/var/spool/cron/crontabs/`
System-wide env vars	`/etc/environment` , `/etc/profile.d/`	`/etc/environment` , `/etc/profile.d/`
Kernel boot parameters	`/etc/default/grub` + `grub2-mkconfig`	`/etc/default/grub` + `update-grub`
Default log directory	`/var/log/` (messages, secure, audit/)	`/var/log/` (syslog, auth.log)

23. Expanded Command Cheat Sheet

Package Management

Task	RHEL 9	Ubuntu 24.04
Install package	`dnf install <pkg>`	`apt install <pkg>`
Remove package	`dnf remove <pkg>`	`apt remove <pkg>`
Update all	`dnf update`	`apt update && apt upgrade`
Search	`dnf search <term>`	`apt search <term>`
List installed	`dnf list installed`	`apt list --installed`
Show package info	`dnf info <pkg>`	`apt show <pkg>`
Clean cache	`dnf clean all`	`apt clean`
Autoremove unused deps	`dnf autoremove`	`apt autoremove`

Service / Process Management

Task	RHEL 9	Ubuntu 24.04
Start/stop/restart service	`systemctl start/stop/restart <svc>`	`systemctl start/stop/restart <svc>`
Enable at boot	`systemctl enable <svc>`	`systemctl enable <svc>`
Check status	`systemctl status <svc>`	`systemctl status <svc>`
View logs for service	`journalctl -u <svc>`	`journalctl -u <svc>`

Networking

Task	RHEL 9	Ubuntu 24.04
Show IP config	`ip a` / `nmcli dev show`	`ip a` / `netplan status`
Set static IP	`nmcli con mod ... && nmcli con up`	Edit `/etc/netplan/*.yaml` then `netplan apply`
Restart networking	`systemctl restart NetworkManager`	`systemctl restart systemd-networkd` / `netplan apply`
DNS lookup/debug	`resolvectl status`	`resolvectl status`

Security

Task	RHEL 9	Ubuntu 24.04
MAC status	`sestatus`	`aa-status`
Set MAC mode	`setenforce 0/1` / `/etc/selinux/config`	`aa-enforce` / `aa-complain <profile>`
Firewall allow port	`firewall-cmd --add-port=80/tcp --permanent && firewall-cmd --reload`	`ufw allow 80/tcp`
Firewall list rules	`firewall-cmd --list-all`	`ufw status verbose`
Check open ports	`ss -tulpn`	`ss -tulpn`

Storage

Task	RHEL 9	Ubuntu 24.04
List block devices	`lsblk`	`lsblk`
Disk usage	`df -h`	`df -h`
LVM volumes	`lvs`, `vgs`, `pvs`	`lvs`, `vgs`, `pvs`
Mount filesystem	`mount /dev/sdX /mnt`	`mount /dev/sdX /mnt`
Filesystem check	`xfs_repair /dev/sdX`	`fsck /dev/sdX`

User Management

Task	RHEL 9	Ubuntu 24.04
Add user	`useradd -m <user>`	`adduser <user>` or `useradd -m`
Add to admin group	`usermod -aG wheel <user>`	`usermod -aG sudo <user>`
Change password	`passwd <user>`	`passwd <user>`

System Info

Task	RHEL 9	Ubuntu 24.04
OS version	`cat /etc/redhat-release`	`lsb_release -a`
Kernel version	`uname -r`	`uname -r`
Subscription/support status	`subscription-manager status`	`pro status`

24. References

- Red Hat Enterprise Linux 9 Documentation — access.redhat.com/documentation
- Red Hat Product Life Cycle Dates — access.redhat.com/support/policy/updates/errata
- Ubuntu 24.04 LTS Release Notes — releases.ubuntu.com/noble
- Ubuntu Release Cycle & Support — ubuntu.com/about/release-cycle
- Ubuntu Pro Documentation — ubuntu.com/pro

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